Rockchip RK3588 Linux6.1 Robot SDK Release Note

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Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian,PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

The document presents the Rockchip RK3588 Linux6.1 Robot SDK release notes, aiming to help engineers get started with the RK3588 Linux6.1 Robot SDK development and related debugging methods more quickly.

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Chipset and System Support

Chipset	Debian
RK3588 Series	Y

Revision History

Date	Version	Author	Revision History
2025-10-20	V0.0.1	WJL	Alpha version.

Contents

Rockchip RK3588 Linux6.1 Robot SDK Release Note

1. Overview
2. Main Functions
   - 2.1 Software Features
   - 2.2 Hardware functions
3. How to Get the SDK
   - 3.1 Get RK3588 Linux6.1 Robot SDK
     - 3.1.1 Get Source Code from Rockchip Code Server
     - 3.1.2 Get Source Code from Local Compression Package
4. Software Development Guide
5. Hardware Development Guide
6. SSH Public Key Operation Introduction
   - 6.1 Key Authority Management
   - 6.2 Reference Documents

1. Overview

The SDK operating system is developed based on Debian 11, supports ROS2 Humble, the kernel is based on Kernel 6.1, which is suitable for the RK3588 series chip development boards and all robot-related products for secondary development based on it.

The RK3588 series chips provide flexible data path combination interfaces to meet customers' freely combined customization needs. For specific function debugging and interface descriptions, please refer to the documents under docs/ in the project directory.

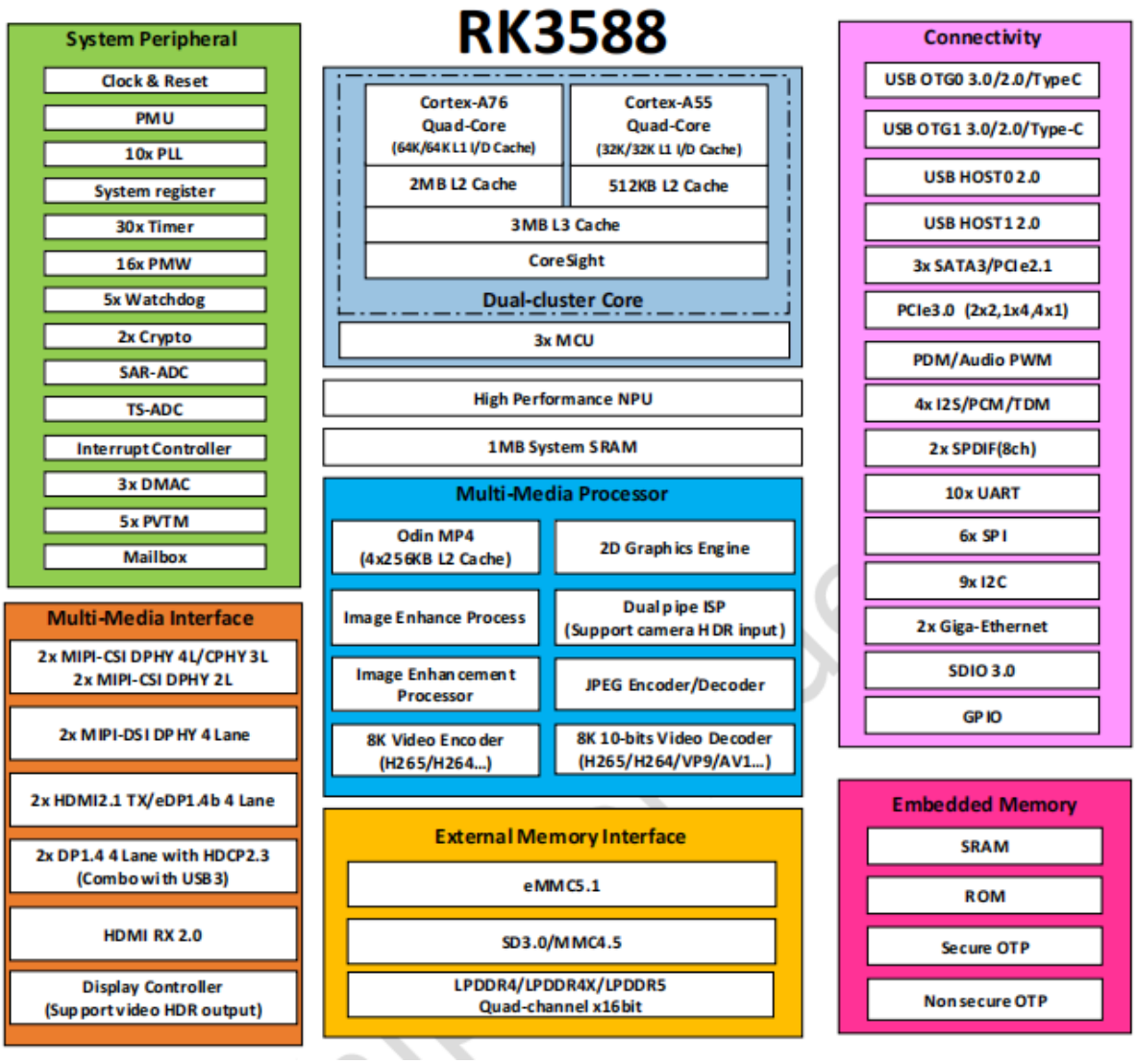
2. Main Functions

2.1 Software Features

Function	Module Name
System	Debian
Partition table	uboot, misc, boot, recovery, rootfs, oem, userdata
File System Type	EXT4
Upgrade Recovery	OTA, Recovery
Secure Boot	SecureBoot
Stress Test Tool	ROCKCHIP_TEST
Data communication	Wi-Fi, Ethernet card, USB, SD card
Applications	ROS2, Multimedia playback, camera preview, browser

2.2 Hardware functions

For specific hardware interface functions, please refer to the RK3588 chip diagram below



3. How to Get the SDK

The SDK is released by Rockchip server. Please refer to Chapter 4 Software Development Guide to build a development environment.

3.1 Get RK3588 Linux6.1 Robot SDK

3.1.1 Get Source Code from Rockchip Code Server

To get RK3588 Linux6.1 Robot SDK software package, customers need an account to access the source code repository provided by Rockchip. In order to be able to obtain code synchronization, please provide SSH public key for server authentication and authorization when apply for SDK from Rockchip technical window. About Rockchip server SSH public key authorization, please refer to Chapter 6 SSH Public Key Operation Introduction.

RK3588 Linux6.1 Robot SDK download command is as follows:

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo \
-u https://gerrit.rock-chips.com:8443/linux/rockchip/rkros/manifests \
-b rk3588 -m rk3588_linux6.1_robot_release.xml
```

Repo, a tool built on Python script by Google to help manage git repositories, is mainly used to download and manage software repository of projects. The download address is as follows:

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

3.1.2 Get Source Code from Local Compression Package

For quick access to SDK source code, Rockchip Technical Window usually provides corresponding version of SDK initial compression package. In this way, developers can get SDK source code through decompressing the initial compression package, which is the same as the one downloaded by repo.
Take RK3588_LINUX6.1_ROBOT_SDK_ALPHA_V0.0.1_20251020.tgz as an example. After geting a initialization package, you can get source code by running the following command:

```
mkdir rk3588_robot
tar xvf RK3588_LINUX6.1_ROBOT_SDK_ALPHA_V0.0.1_20251020.tgz -C rk3588_robot
cd rk3588_robot
.repo/repo/repo sync -l
.repo/repo/repo sync -c
```

Developers can update via `.repo/repo/repo sync -c` command according to update introductions that are regularly released by FAE window.
If you encounter issues downloading the repository, you can use the `--force-sync` parameter to force an update, as in `.repo/repo/repo sync -c --force-sync`. Before doing so, please ensure that any local modifications have been backed up.

After updating the SDK code, a clean operation is required. For example: `./build.sh cleanall`.

> Note:
> The software release version can be viewed through engineering XML, as follows:

```
.repo/manifests$ realpath rk3588_linux6.1_robot_release.xml
# e.g.:printf version v0.0.1, update time on 20251020
<SDK>/.repo/manifests/release/rk3588_linux6.1_robot_alpha_v0.0.1_20251020.xml
```

4. Software Development Guide

For software development, please refer to the quick start documents in the project directory:

```
<SDK>/robot/docs/en/RK3588/Quick-start/
└── Rockchip_RK3588_Quick_Start_Linux_Robot_EN.pdf

<SDK>/docs/en/RK3588/Quick-start/
└── Rockchip_RK3588_Quick_Start_Linux_EN.pdf
```

5. Hardware Development Guide

For hardware development, please refer to the user guide document in the project directory:

```
<SDK>/docs/en/RK3588/Hardware/
...
├── Rockchip_RK3588_EVB1_User_Guide_V1.2_EN.pdf
├── Rockchip_RK3588_EVB7_User_Guide_V1.0_EN.pdf
└── Rockchip_RK3588_Hardware_Design_Guide_V1.4_EN.pdf
```

6. SSH Public Key Operation Introduction

Please follow the introduction in the
Rockchip_User_Guide_SDK_Application_And_Synchronization_EN to generate an SSH public key and send the email to fae@rock-chips.com, to get the SDK code.
This document will be released to customers during the process of applying for permission.

6.1 Key Authority Management

Server can monitor download times and IP information of a key in real time. If an abnormality is found, download permission of the corresponding key will be disabled.
Keep the private key file properly. Do not grant second authorization to third parties.

6.2 Reference Documents

For more details, please refer to document:

```
<SDK>/docs/en/Others/
└── Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf
```